FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2022

TEST DATE: 17-02-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

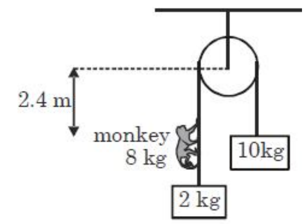
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

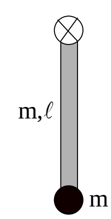
1. Two blocks of mass 10 kg and 2 kg respectively are connected by an ideal string passing over a fixed smooth pulley as shown in figure. A monkey of 8 kg started climbing the string with constant acceleration $2ms^{-2}$ with respect to string at t = 0. Initially the monkey is 2.4 m from the pulley. Find the time taken by the monkey to reach the pulley.



(A) 1 sec
(B) 2 sec
(C) 4 sec
(D) 8 sec

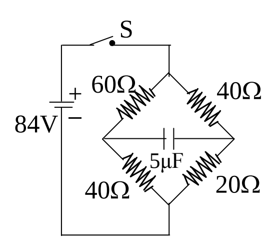
2. Which of the following transitions of He^{+} ion will give rise to spectral line which has same wavelength as some spectral line in hydrogen atom?
(A) $n=4$ to $n=2$
(B) $n=6$ to $n=5$
(C) $n=6$ to $n=3$
(D) None of these

3. A particle is attached to the lower end of a uniform rod which is hinged at its other end as shown in the figure. Another identical particle moving horizontally collides inelastically and sticks to it. The minimum speed of moving particle so that the rod with particles performs circular motion in a vertical plane will be: [length of the rod is ℓ, consider masses of both particles and rod to be same]



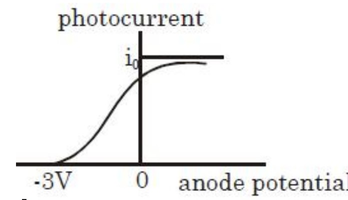
(A) $\sqrt{20g\ell}$
(B) $\sqrt{10g\ell}$
(C) $\sqrt{70/3g\ell}$
(D) $\sqrt{175/3g\ell}$

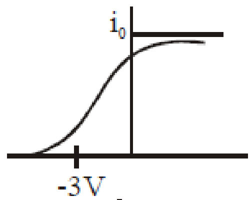
4. The circuit shown in the figure is in steady state for a long time. The connection to battery is suddenly broken (switch S is opened up). What is the charge (in μC) on the capacitor after 0.001 sec.?

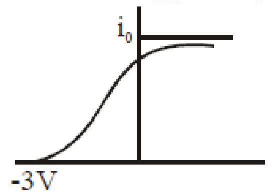


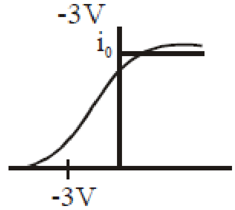
(A) 100
(B) $100e^{-2}$
(C) $100e^{-4}$
(D) 0

5. Figure shows the anode potential vs photo-current graph when photons of 6 eV are incident on cathode in a photoelectric experiment set-up. In the same experimental set-up, if photons of 8 eV and same intensity are incident on cathode, then anode-potential vs photo-current graph will be (assume 100% efficiency of photons to eject photo-electrons in both cases)



(A)


(B)


(C)


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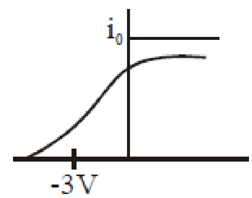
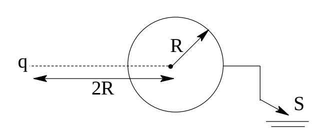


6. A sound source is located somewhere along the x-axis. Experiment shows that the same wavefront simultaneously reaches listeners at x = –10m and x = +4.0m. A third listener is positioned along the positive y-axis. What is y-coordinate (in m) if the same wavefront reaches at him at the same instant it does the first two listeners?

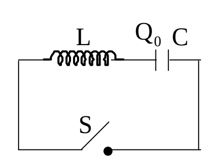
(A) $\sqrt{40}$
(B) 3
(C) 7
(D) $\sqrt{30}$

7. A point charge is kept outside a conducting sphere as shown in figure. The charge flown through the switch in to the earth after closing the switch is:



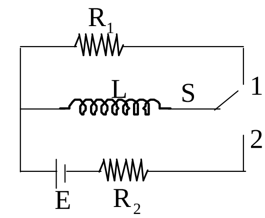
(A) $-\frac{q}{2}$
(B) $\frac{q}{2}$
(C) $-\frac{q}{4}$
(D) $\frac{q}{4}$

8. At t < 0, the capacitor is charged and the switch is opened. At t = 0 the switch is closed. The shortest time T at which the charge on the capacitor will be zero is given by:



(A) $\pi\sqrt{LC}$
(B) $\frac{3}{2}\pi\sqrt{LC}$
(C) $\frac{\pi}{2}\sqrt{LC}$
(D) $2\pi\sqrt{LC}$

9. In the circuit shown switch S is connected to position 2 for a long time and then joined to position 1. The total heat produced in resistance R_1 is:



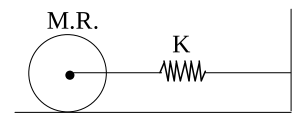
(A) $\frac{LE^2}{2R_2^2}$

(B) $\frac{LE^2}{2R_1^2}$

(C) $\frac{LE^2}{2R_1R_2}$

(D) $\frac{LE^2\left(R_1+R_2\right)^2}{2R_1^2R_2{}^2}$

10. A solid cylinder attached to horizontal massless spring can roll without slipping along horizontal surface. Find time period of oscillation:



(A) $2\pi\sqrt{M/2k}$

(B) $\pi\sqrt{3M/2k}$

(C) $\pi\sqrt{2M/3k}$

(D) $2\pi\sqrt{3M/2k}$

11. Assuming the sun to be a spherical body (e = 1) of radius R at a temperature of T K, evaluate the total radiant power, incident on Earth having radius r_0, at a distance r from the Sun, where r_0 is the radius of the earth and σ is Stefan's constant.

(A) $\frac{\pi r_0^2R^2\sigma T^4}{r^2}$

(B) $\frac{r_0^2R^2\sigma T^4}{4\pi r^2}$

(C) $\frac{R^2\sigma T^4}{r^2}$

(D) $\frac{4\pi r_0^2R^2\sigma T^4}{r^2}$

12. The minimum and maximum distance of a satellite from the centre of the earth are 2R and 4R, respectively, where R is the radius of earth and M is the mass of earth. The radius of curvature at the point of maximum distance is:

(A) 8R/3

(B) 4R/3

(C) 3R/8

(D) 3R/4

13. A man standing on a road has to hold his umbrella at 30° with the vertical to keep the rain away. He throws the umbrella and starts running at 10km/h. He finds that raindrops are hitting his head vertically. The actual speed of rain drops is
(A) $20\,\text{km/h}$
(B) $10\sqrt{3}\,\text{km/h}$
(C) $20\sqrt{3}\,\text{km/h}$
(D) $10\,\text{km/h}$

14. 70 cal of heat is required to raise the temperature of 2 moles of an ideal gas from 30°C to 35°C while the pressure of the gas is kept constant. The amount of the heat required to raise the temperature of the same gas through the same temperature range at constant volume is (gas constant, R = 2 cal $\text{mol}^{-1}\text{K}^{-1}$)
(A) 80 cal
(B) 70 cal
(C) 60 cal
(D) 50 cal

15. In a Lloyd's mirror experiment, a light wave emitted directly by the source S interferes with reflected light from the mirror. The screen is 1m away from the source S. The size of the fringe width is 0.25mm. The source is moved 0.6mm above the initial position, the fringe width decreases by 1.5 times. The light used has wavelength
(A) 0.6 μm
(B) 0.4 μm
(C) 0.2 μm
(D) 0.8 μm

16. There is a simple pendulum hanging from ceiling of a lift. When the lift is stand still, the time period of the pendulum is T. If the lift moving upward with deceleration 3g/4, where g is acceleration due to gravity, then the new time period of pendulum is:
(A) 0.75 T
(B) 0.25 T
(C) 2T
(D) 4T

17. A solid sphere of mass 1 kg, radius 10 cm rolls down an inclined plane of height 7m. The velocity of its centre as it reaches the ground level is:
(A) 7 m/s
(B) 10 m/s
(C) 15 m/s
(D) 20 m/s

18. Two spheres made of same material have radii in the ratio $2:1$. If both the spheres are at same temperature, then what is the ratio of heat radiation energy emitted per second by them?
(A) $1:4$
(B) $4:1$
(C) $3:4$
(D) $4:3$

19. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of
    (A) 1000
    (B) 10,000
    (C) 10
    (D) 100

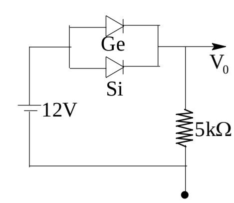
20. Lenses of power 3D and –5D are combined to form a compound lens. An object is placed at a distance of 50 cm from this lens. Its image will be formed at a distance from the lens will be
    (A) 25 cm
    (B) 20 cm
    (C) 30 cm
    (D) 40 cm

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A plastic circular disc of radius 10 cm is placed on a thin oil film of thickness 2 mm, spread over a flat horizontal surface. The torque (N-m) required to spin the disc about its central vertical axis with a constant angular velocity 8 rad/sec is (coefficient of viscosity of oil is 1.0 kg/m-s) (take π = 3.14)

22. Calculate the value of output voltage V_0 (in V) if the Si diode and Ge diode conduct at 0.7 V and 0.3V respectively as shown in figure.



23. In Young's double slit experiment, the wavelength of red light is 7800 $\mathring{A}$ and that of blue light is 5200 $\mathring{A}$. The minimum value of n for which n^{th} bright band due to red light coincides with $(n+1)^{th}$ bright band due to blue light, is:

24. A railway track (made of iron) is laid in winter when the average temperature is 18°C. The track consists of sections of 12.0 m placed on after the other. How much gap (in cm) should be left between two such sections so that there is no compression during summer when the maximum temperature goes to 48°C? Coefficient of linear expansion of iron $=11\times10^{-6}$ / °C_____

25. A rod is clamped at both of its ends and stationary longitudinal waves are produced in it. In first experiment, the rod has a total of 4 nodes with amplitude of each anti-node to be 1 mm. In other experiment, the rod has a total of 6 nodes with amplitude of each anti node to be 2mm. The energy of vibrations in the rod in two cases is E_1 and E_2 respectively. Write value of $\frac{E_1}{E_2}$.

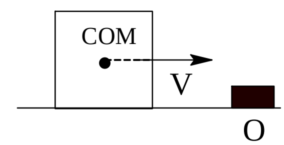
26. A system of binary stars of masses M_1 and M_2 are moving in circular orbits of radii r_1 and r_2 respectively. If T_1 and T_2 are the time periods of revolution of masses M_1 and M_2 respectively and if $\frac{r_1}{r_2}=2$, then find $\frac{T_1}{T_2}$.

27. A bullet of mass 10 gm is fired horizontally with velocity 1000 m/s from rifle situated at height 50m above the ground. If the bullet reaches the ground with velocity 500 m/s, find work done against air resistance in the trajectory of the bullet in kilo-joule.

28. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequency of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency (in Hz) for this string is $100x$. Find x.

29. A boat is moving due east in a region where the earth's magnetic field is $5.0\times10^{-5}\mathrm{NA^{-1}m^{-1}}$ due north and horizontal. The boat carries a vertical aerial 3m long. If the speed of the boat is 3m/s, find the magnitude of the induced emf in the wire of aerial (in mV).

30. An ice cube of mass M and with sides of length a is sliding without friction with speed v_0 when it hits a ridge O at the edge of counter (see figure). This collision causes the cube to tilt as shown. The minimum value of v_0 needed for the cube to fall off the table is given by $\sqrt{kag}$. Find value of k.



Chemistry

PART – B

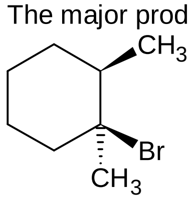
SECTION – A
(One Options Correct Type)

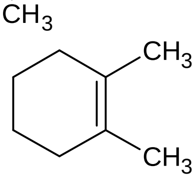
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

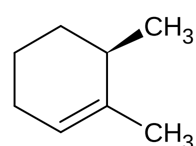
31. Which of the following statement is incorrect?
   (A) H^+ is the smallest size cation in the periodic table
   (B) Van der waals radius of chlorine is more than covalent radius
   (C) Ionic mobility of hydrated Li^+ is greater than that of hydrated Na^+
   (D) He atom has the highest ionisation enthalpy in the periodic table

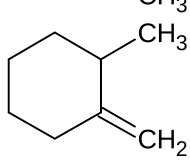
32. Identify the statement(s) which is not correct with respect to surface phenomenon.
   (A) If on adding electrolyte in an emulsion, the conductivity increases then it will be oil in water type emulsion.
   (B) Tyndall effect is observed when refractive indices of dispersed phase and dispersion medium differ largely.
   (C) Macromolecular colloids are generally lyophobic in nature
   (D) Gases which can react with adsorbents are generally chemisorbed

33. The major product obtained when following substrate is subjected to $E2$ reaction will be:

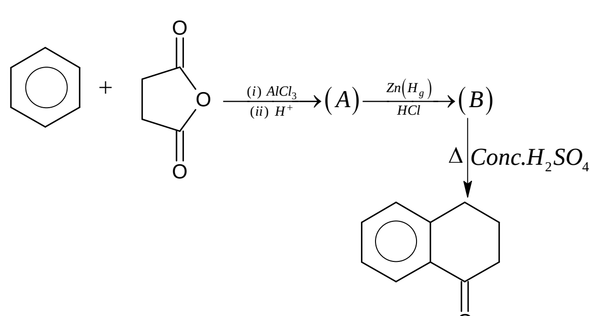


(A)


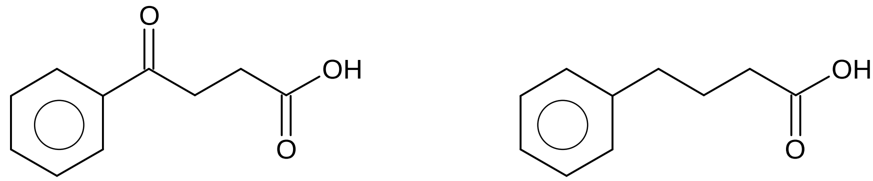
(B)


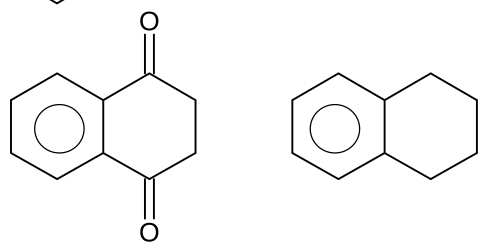
(C)


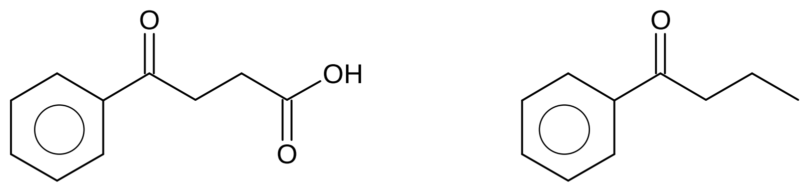
(D) None of these

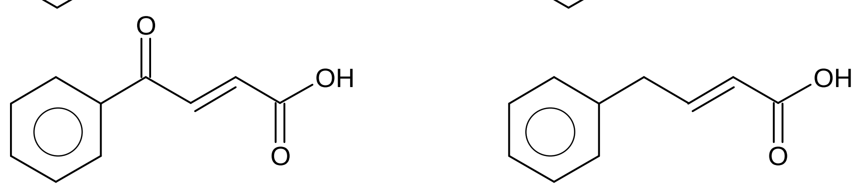
34.


Structure of (A) and (B) respectively are:

(A) 

(B) 

(C) 

(D) 

35. An aqueous solution of a metal ion (A) on reaction with KI gives a black brownish ppt (B) and this aqueous suspension on treatment with excess KI gives orange yellowish solution. Then A is
(A) pb^{2+}
(B) Bi^{3+}
(C) Hg_2^{2+}
(D) Hg^{2+}

36. Which of the following compounds consist of a P – P linkage?
(A) Hypo phosphoric acid
(B) Pyrophosphorous acid
(C) Dipolyphosphoric acid
(D) Metaphosphoric acid

37. The specific conductance of a saturated solution of silver bromide is $\mathrm{K\,Scm^{-1}}$. The limiting ionic conductivity of $\mathrm{Ag^+}$ and $\mathrm{Br^-}$ ions are x and y respectively. The solubility of silver bromide in $\mathrm{g/L}$ is (molar mass of $\mathrm{AgBr} = 188$)
(A) $\dfrac{K \times 1000}{x - y}$
(B) $\dfrac{K}{x + y} \times 188$
(C) $\dfrac{K \times 1000 \times 188}{x + y}$
(D) $\dfrac{x + y}{k} \times \dfrac{1000}{188}$

38. A certain amount of reducing agent reduces x mole of $\mathrm{KMnO_4}$ & y mole of $\mathrm{K_2Cr_2O_7}$ in different experiments in acidic medium. If the change in oxidation state of reducing agent is same in both experiments then $x : y$ is.
(A) $5:3$
(B) $3:5$
(C) $5:6$
(D) $6:5$

39. The activation energies of two reactions are $\mathrm{E_1}$ & $\mathrm{E_2}$ with $\mathrm{E_1 > E_2}$. If temperature of reacting system is increased from T_1 (rate constant are $\mathrm{k_1}$ and $\mathrm{k_2}$) to T_2 (rate constant are $\mathrm{k_1^1}$ and $\mathrm{k_2^1}$) predict which of the following alternative is correct.
(A) $\frac{k_1^1}{k_1} = \frac{k_2^1}{k_2}$
(B) $\frac{k_1^1}{k_1} > \frac{k_2^1}{k_2}$
(C) $\frac{k_1^1}{k_1} < \frac{k_2^1}{k_2}$
(D) $k_1^1 < k_2^1$

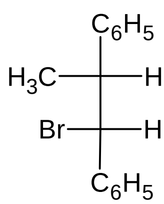
40. The wavelength of the first Lyman lines of hydrogen, He^+ and Li^{2+} ions and λ_1, λ_2 & λ_3. The ration of these wavelengths is
(A) $1:4:9$
(B) $9:4:1$
(C) $36:9:4$
(D) $6:3:2$

41. Two system $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ and $COCl(g) \rightleftharpoons CO(g) + Cl_2(g)$ are simultaneously in equilibrium in a vessel at constant volume. If some CO is introduced in the vessel, then at new equilibrium, the modes of
(A) PCl_5 increase
(B) PCl_5 remain unchanged
(C) PCl_5 decrease
(D) Cl_2 increase

42. In the following statement, which combination of true (T) and false (F) options in correct?
(A) Ionic mobility is highest for I^- in water as compared to other halides.
(B) IF_5 is square pyramidal and IF_7 is pentagonal bipyramidal in shape
(C) Reactivity order is $F_2 < Cl_2 < Br_2 < I_2$
(D) Oxidising power order is $F_2 < Cl_2 < Br_2 < I_2$
(A) TFTF
(B) TTFT
(C) TTTT
(D) TTFF

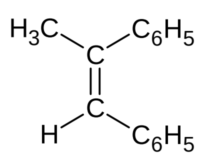
43. Xenon fluorides are very good oxidising and fluorinating agents. They also act as F^- donors and acceptors. When XeF_4 donates its fluoride to SbF_5, then the states of hybridisation of central atom of cationic part and anionic part of product are:
(A) sp^3d, sp^3d^2
(B) sp^3d^2, sp^3d
(C) sp^3d^2, sp^3d^2
(D) dsp^2, sp^3

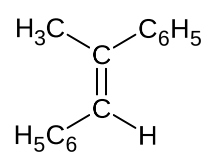
44. Which of the following is a polar molecule?
(A) XeF_4
(B) BF_3
(C) C_2F_4
(D) PCl_2F_3

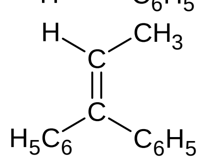
45. 

$\xrightarrow[E2]{alcoholic\ KoH} A$

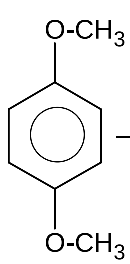
A is

(A) 

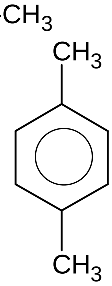
(B) 

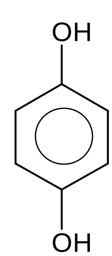
(C) 

(D) None of these

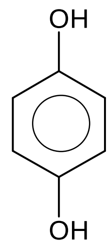
46. 

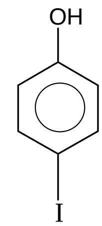
$\xrightarrow[(excess)]{Conc.HI}$ Products, Products of reaction are

(A) 

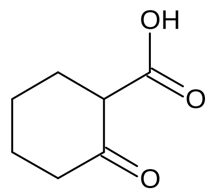
$+2CH_3I$
(B) 

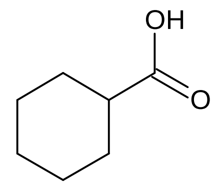
$+2CH_3OH$

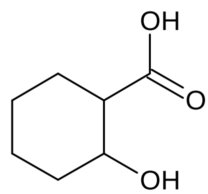
(C)  $+2CH_3I$

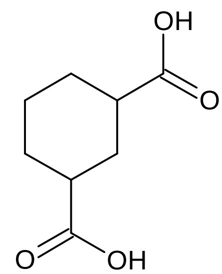
(D)  $+2CH_3I$

47. Which of the following compound loses CO_2 upon heating to $100°C$

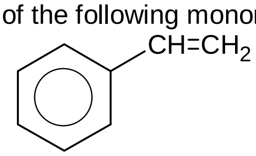
(A) 

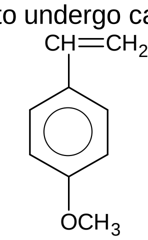
(B) 

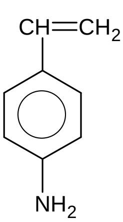
(C) 

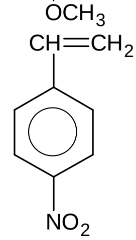
(D) 

48. Which of the following monomers has greatest ability to undergo cationic polymerisation?

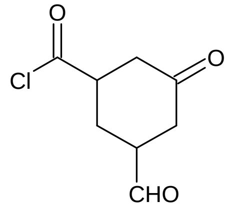
(A) 

(B) 

(C) 

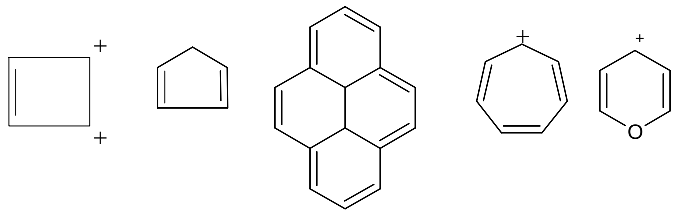
(D) 

49. How many molecules of $RMgX$ are consumed in the below reaction

 $\xrightarrow[H_3O^+]{RMgX}$

(A) 2
(B) 4
(C) 5
(D) 6

50. Amongst the following, total no of aromatic compounds are



(A) 5
(B) 4
(C) 3
(D) 6

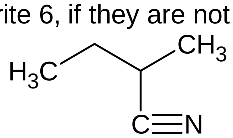
SECTION – B
(Numerical Answer Type)

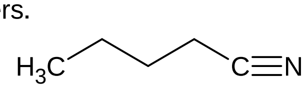
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

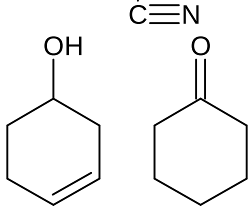
51. One mole of calcium phosphide on reaction with excess of water gives how many moles of phosphine.

52. Three moles of an ideal gas $\left(C_{p,m}=2.5R\right)$ and 2 moles of another ideal gas $\left(C_{p,m}=3.5R\right)$ are taken in a vessel and compressed reversible and adiabatically. In this process, the temperature of gaseous mixture increased from 300K to 400K. The increase in internal energy of gaseous mixture (in K cal) is. [R = 2 cal/mol/K]

53. Analyse the following pairs of compounds.
Write 1, if they are Tautomers.
Write 2. if they are Metamers.
Write 3, if they are Position isomers.
Write 4, if they are Functional group isomers.
Write 5, if they are Chain isomers.
Write 6, if they are not isomers.

(a) 



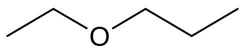
(b) 

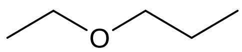
(c) $CH_3-CH_2-O-CH_2-CH_3$
$CH_3-CH_2-CH_2-O-CH_3$

Write answer of part (a), (b), (c) in the same order and present the three digit number and report answer in sum of digit.

54. When mixture of $NaCl$ and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4 then compound X is formed. What is the oxidation state of central atom of X.

55. Sum of molecular mass of alkyl iodides produced in following reaction is (At weight of Iodine=127)

(a)  $\xrightarrow{\text{conc.HI}}$ alcohol+alkyl iodide

(b)  $\xrightarrow{\text{anhydrous}}$ alcohol+alkyl iodide

(c) Ph - O - Me $\xrightarrow[\text{excess}]{\text{HI}}$ alcohol+alkyl iodide

56. Assuming that boron atoms arrange themselves at the corners of truncated octahedron. Find the $|\Delta H|$ (in kJ/mol) if 1 mol truncated octahedron is formed by required boron atoms in gaseous state. Given: $\Delta H_{BE}(B\text{-}B) = 200\,kJ/mol$

[Write your answer as sum of digits till you get the single digit answer].

57. Find the number of species from the following which has magnetic moment value of 1.73 B.M.

$Fe^{2+}, Cu^{2+}, Ni^{2+}, NO_2, NO_2, Sc^{2+}$

58. For any sparingly soluble salt $\left[M(NH_3)_4Br_2\right]H_2PO_2$

Given: $\lambda^0_{M(NH_3)_4Br_2^+} = 400\,Sm^2mol^{-1}$, $\lambda^0_{H_2PO_2^-} = 100\,Sm^2mol^{-1}$

Specific resistance of saturated $\left[M(NH_3)_4Br_2\right]H_2PO_2$ solution is $200\,\Omega\,cm$. If solubility product constant of the above salt is 10^{-x}. What will be the value of x.

59. How many among the following is / are antipyretic drugs
Ibuprofen, Codeine, Nicotine, Aspirin, Chloroquine, phenacitin, Naproxen

60. How many of the following reagent/s will liberate at least one oxide of nitrogen as a product?
(i) Ag + conc. HNO_3
(ii) Sn + dil., HNO_3 (20%)
(iii) Cu + conc. HNO_3
(iv) C + conc. HNO_3
(v) Zn + conc. HNO_3
(vi) Zn + dil. HNO_3 (20%)
(vii) P_4 + conc. HNO_3
(viii) S_8 + conc. HNO_3
(ix) Cu + dil. HNO_3 (20%)

Mathematics PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. $\sin 47° - \sin 25° + \sin 61° - \sin 11° =$
(A) $\cos 7°$
(B) $\sin 7°$
(C) $2\cos 7°$
(D) $2\sin 7°$

62. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T'. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N'. The areas of the triangles PNT and PN'T' are Δ and Δ' respectively, then $\frac{1}{\Delta} + \frac{1}{\Delta'}$ is
(A) equal to 1
(B) depends on t
(C) depends on c
(D) equal to 2

63. The normal at P to a hyperbola of eccentricity e, intersects its transverse and conjugate axes at L and M respectively. If locus of the mid-point of LM is hyperbola, then eccentricity of the hyperbola is:
(A) $\left(\frac{e+1}{e-1}\right)$
(B) $\frac{e}{\sqrt{(e^2-1)}}$
(C) e
(D) None of these

64. Area of the triangle formed by the lines $x - y = 0, x + y = 0$ and any tangent to the hyperbola $x^2 - y^2 = a^2$ is:
(A) $|a|$
(B) $\frac{1}{2}|a|$
(C) a^2
(D) $\frac{1}{2}a^2$

65. The point, at shortest distance from the line $x+y=7$ and lying on an ellipse $x^2+2y^2=6$, has co-ordinates:

(A) $\left(\sqrt{2},\sqrt{2}\right)$

(B) $\left(0,\sqrt{3}\right)$

(C) $(2,1)$

(D) $\left(\sqrt{5},\dfrac{1}{\sqrt{2}}\right)$

66. Locus of the foot of perpendicular drawn from origin to any tangent to the ellipse $\dfrac{x^2}{2}+y^2=1$ is:

(A) $\left(x^2+y^2\right)^2=2x^2+y^2$

(B) $\left(x^2-y^2\right)^2=2x^2+y^2$

(C) $\left(x^2+y^2\right)^2=2x^2-y^2$

(D) None of these

67. For parabola $x^2+y^2+2xy-6x-2y+3=0$, then focus is:

(A) $(1,-1)$

(B) $(-1,1)$

(C) $(3,1)$

(D) None of these

68. The condition that the parabola $y^2=4c\left(x-d\right) and\ y^2=4ax$ have a common normal other than x- axis $\left(a>0,c>0\right)$ is:

(A) $2a<2c+d$

(B) $2c<2a+d$

(C) $2d<2a+c$

(D) $2d<2c+a$

69. The point $\left([P+1],[P]\right)$, (where $[.]$ denotes the greatest integer function) lying inside the region bounded by the circle $x^2+y^2-2x-15=0\ and\ x^2+y^2-2x-7=0$ then:

(A) $p\in[-1,0)\cup[0,1)\cup[1,2)$

(B) $p\in[-1,2)-\{0,1\}$

(C) $p\in(-1,2)$

(D) None of these

70. If the distance from the origin of the centres of three, circles $x^2+y^2+2\lambda_i x-c^2=0,(i=1,2,3)$ are in GP. then the lengths of the tangents drawn to them from any point on the circle $x^2+y^2=c^2$ are in:
(A) A. P
(B) G. P
(C) H.P
(D) None of these

71. The values of k for which the circle $x^2+y^2=1 \, and \, x^2+y^2-4\lambda x+8=0$ have two common tangents is:
(A) $\left(-\frac{9}{4},\frac{9}{4}\right)$
(B) $\left(-\infty,-\frac{9}{4}\right]\cup\left[\frac{9}{4},\infty\right)$
(C) $\left(-\infty,-\frac{9}{4}\right)\cup\left(\frac{9}{4},\infty\right)$
(D) None of these

72. The equation of image of pair of lines $y=|x-1|$ in y- axis is:
(A) $x^2+y^2+2x+1=0$
(B) $x^2-y^2+2x-1=0$
(C) $x^2-y^2+2x+1=0$
(D) None of these

73. If $\alpha,\beta>0 \, and \, \alpha<\beta \, and \, ax^2+4\gamma xy+\beta y^2+4p(x+y+1)=0$ represent a pair of straight lines, then:
(A) $\alpha\le p\le\beta$
(B) $p\le\alpha$
(C) $p\ge\alpha$
(D) $p\le\alpha \, or \, p\ge\beta$

74. If $\frac{2}{1!9!}+\frac{2}{3!7!}+\frac{4}{5!5!}=\frac{2^m}{n!}$, then orthocentre of the triangle having sides $x-y+1=0, x+y+3=0 \, and \, 2x+5y-2=0$ is:
(A) $(2m-2n,m-n)$
(B) $(2m-2n,n-m)$
(C) $(2m-n,m+n)$
(D) $(2m-n,m-n)$

75. In ΔABC, if $p=\tan\frac{B-C}{2}\cot\frac{B+C}{2}$ $q=\tan\frac{C-A}{2}\cot\frac{C+A}{2}$ and $r=\tan\frac{A-B}{2}\cot\frac{A+B}{2}$ then $\sum p=$:
(A) 1
(B) $\sum pq$
(C) $-pqr$
(D) None of these

76. If I be the incentre of ΔABC and P,Q are feet of perpendicular from A to BI,CI respectively. Then $AQ:CI+AP:BI=$:
(A) $\cot\left(\frac{A+B}{2}\right)$
(B) $\cot\left(\frac{B+C}{2}\right)$
(C) $\cot\left(\frac{C+A}{2}\right)$
(D) $\tan\left(\frac{B+C}{2}\right)$

77. A,B,C are three points on a horizontal line through the base O of a pillar OP, such that OA, OB, OC are in A. P. If α,β,γ the angles of elevation of the top of the pillar at A,B,C respectively are also in A. P. then $\sin\alpha,\sin\beta,\sin\gamma$ are in:
(A) A. P.
(B) G. P.
(C) H. P.
(D) None of these

78. If $n\in N,$ then $7^{2n}+2^{3n-3}.3^{n-1}$ is always divisible by:
(A) 25
(B) 35
(C) 45
(D) None of these

79. Let $P(n)$ denotes the statement that n^2+n is odd. It is seen that $P(n)\Rightarrow P(n+1).P_n$ is true for all:
(A) $n>1$
(B) $n\in N$
(C) $n>2$
(D) None of these

80. If $\sin^{-1}\left(\tan\left(\frac{\pi}{2}\sin\theta\right)\right)+\cos^{-1}\left(\cot\left(\frac{\pi}{2}\cos\theta\right)\right)=\frac{\pi}{2}$, then value of $\sin\frac{\theta}{3}$ can be:
(A) 0
(B) $\frac{1}{2}$
(C) $\frac{1}{\sqrt{2}}$
(D) None of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. The number of ordered pairs (x, y) of real numbers satisfying $4x^2-4x+2=\sin^2 y$ and $x^2+y^2\le 3$ is equal to___

82. A rectangular hyperbola whose centre is C is cut by any circle of radius unity in four points P, Q, R and S. Then $CP^2+CQ^2+CR^2+CS^2$ is equal to____

83. The line $5x-3y=8\sqrt{2}$ is a normal to the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. If $\frac{\pi}{\mathrm{k}}$ be the eccentric angle of the foot of this normal, then k is_____

84. The triangle formed by the tangent to the parabola $y=x^2$ at the point whose abscissa is $x_0\ (x_0\in[1,2])$, the y-axis and the straight line $y=x_0^2$ has the greatest area if x_0 is equal to__

85. The polynomial R(x) is the remainder upon dividing x^{2007} by x^2-5x+6. If R (0) can be expressed as $ab(a^c-b^c)$, find the value of $\frac{\mathrm{a+b}}{20}+\mathrm{c}=?$

86. Number of roots of the equation $\cos^2 x+\left(\frac{\sqrt{3}+1}{2}\right)\sin x-\frac{\sqrt{3}}{4}-1=0$ which lie in the interval $[-\pi,\pi]$ is_____

87. The number of solutions of $\tan(5\pi\cos\theta)=\cot(5\pi\sin\theta)$ for $\theta\in(0,2\pi)$ is_____

88. Number of solutions of the equation $4\sin^2 x+\tan^2 x+\cot^2 x+\operatorname{cosec}^2 x=6$ in $[0,\pi]$ is___

89. A special deck of cards contains 49 cards, each labeled with a number from 1 to 7 and coloured with one of seven colours. Each number – colour combination appears on exactly one card. Mokshika will select a set of eight cards from the deck at random. Given that she gets atleast one card of each colour and atleast one card with each number, the probability that Mokshika can discard one of her cards and still have atleast one card of each colour and atleast one card with each number is $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $\left|p-q\right|$:

90. If $f\left(y\right)=\lim_{n\to\infty}\sum_{n=1}^{n}\frac{e^{n+y}\left(e-1\right)}{\left(e^{n}-1\right)\left(e^{n+1}-1\right)}$ then the value of $\int_{0}^{1} f\left(x\right)dx$ is: